Green Technologies and Sustainability 4 (2026) 100352

Contents lists available at ScienceDirect

Green Technologies and Sustainability

journal homepage:
https://www.keaipublishing.com/en/journals/green-technologies-and-sustainability/



Full-length article

Environmental impact analysis of structural steels using retrieval-augmented generation and large language models



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HIGHLIGHTS

- EcoQuery automates LCA data use from steel EPDs.
- RAG-LLM improves accessibility of carbon data in design.
- Reduces barriers in life cycle sustainability assessment.
- Supports low-carbon decisions in the built environment.
- Scalable AI tool for cross-sector environmental impact analysis.

ARTICLE INFO

Dataset link: https://www.eco-platform.org/home.html, https://eco-portal.eco-platform.org/

Keywords:
Life Cycle Assessment (LCA)
Low-carbon technologies
Sustainable structural materials

ABSTRACT

Reducing embodied carbon in the built environment requires reliable methods to assess structural materials, particularly steel, one of the most carbon-intensive. While environmental product declarations (EPDs) provide standardised and verified life cycle data, their integration into design workflows remains limited due to challenges in accessibility, interpretation, and cross-product comparison. Moreover, the growing volume and complexity of EPDs demand significant technical expertise for effective use with analytical tools. This study introduces EcoQuery, an artificial intelligence (AI) assistant based on a retrieval-augmented generation (RAG) framework integrated with large language models (LLMs), designed to address these barriers. EcoQuery enables natural language (NL) question answering (QA) and interpretation of EPD data for structural steel products by automating the extraction of environmental indicators and supporting reasoning tasks. EcoQuery demonstrated strong performance, achieving 83% accuracy, 87% precision, 85% recall, 81% exact match, and a Bidirectional Encoder Representations from Transformers Score (BERTScore) of 0.89, indicating high semantic fidelity. By using RAG to connect EPD data with language model reasoning, EcoQuery lowers the expertise required to work with life cycle data and supports carbon-conscious, data-driven material decisions. While demonstrated on steel EPDs, EcoQuery is adaptable across materials and sectors, offering a scalable pathway to decarbonising the built environment.

1. Introduction

Buildings are among the largest contributors to global emissions, accounting for approximately 39% of annual carbon dioxide (CO_2) output [1]. Among the materials used in construction, structural steel plays a central role, widely used in buildings and infrastructure, but also a major source of embodied carbon [2,3]. This makes structural steel a high-impact target for emissions reduction, particularly during early design stages where material choices significantly influence whole-life carbon outcomes. Steel presents both challenges and opportunities in this context [4]. Its production is energy-intensive and often relies on fossil fuels [5], resulting in high global warming potential (GWP) and abiotic depletion potential of fossil fuels (ADPF). However, it is also one of the most recyclable construction materials [6], and modern production methods increasingly use scrap content. When properly accounted for in life cycle assessments (LCAs), these recycling benefits can significantly reduce the net environmental impacts. As environmental regulations evolve, there is a growing requirement to disclose the environmental effects of construction materials [7]. This makes EPDs critical for compliance and transparency [8]. Defined by International Organization for Standardisation (ISO) [9], EPDs are standardised

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https://doi.org/10.1016/j.grets.2026.100352
Received 26 October 2025; Received in revised form 18 January 2026; Accepted 2 February 2026
Available online 4 February 2026
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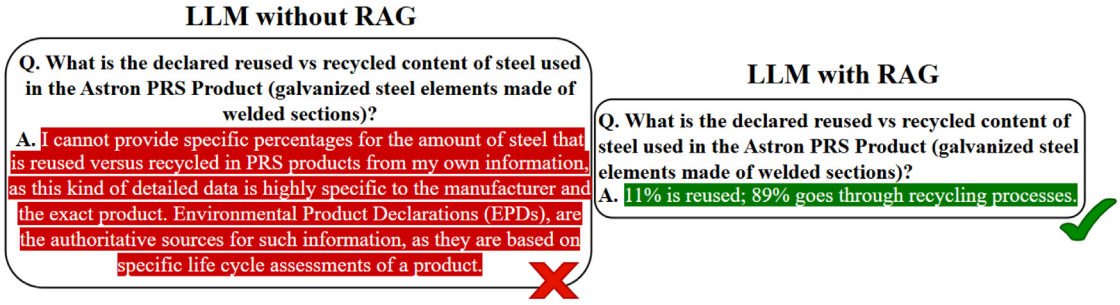


Fig. 1. Impact of RAG on LLM response accuracy.

documents that present scientifically grounded, quantified environmental data based on LCAs. Covering every stage of a product's lifecycle, from raw material extraction to manufacturing, use, and end-of-life, they enable fair comparisons between materials serving the same purpose.

In the construction sector, EPDs are essential for reducing embodied carbon and advancing broader sustainability goals [10]. They support architects, engineers, and designers in early-stage material selection by highlighting environmentally preferable options, while enabling manufacturers to assess product performance and demonstrate carbon transparency. For whole life carbon assessments (WLCA), steel EPDs play a critical role in quantifying the full environmental footprint of buildings and informing low-carbon design decisions [11]. Despite growing availability through platforms like the ECO-Platform [12] and national databases, EPDs remain underutilised in mainstream design workflows [13]. This gap is driven by challenges such as inconsistent data formats, limited standardisation, and complex integration with digital design tools [14]. Many EPDs are published as semi-structured international life cycle data (ILCD) PDFs, requiring specialised knowledge to interpret life cycle stages, impact indicators, and methodological assumptions [15]. Additionally, existing portals often rely on fragmented search tools or token-restricted application programming interfaces (APIs), making timely access difficult for practitioners during early project phases. Given the pivotal role of structural steel in decarbonising the built environment, improving the usability, standardisation, and digital integration of EPD data is essential. This will enable data-driven, low-carbon material choices across the construction industry.

To use the full potential of EPDs in early-stage decision-making, advanced AI-driven methods [16] remain underexplored but offer significant opportunities for innovation and impact. Among these, LLMs, a class of AI systems trained on vast text datasets, excel at understanding, generating, and interpreting human language [17]. They are effective at tasks like summarisation and QA but are limited by their training data [18], often lacking access to domain-specific sources like EPDs. As a result, standalone LLMs frequently return generic or partial answers when queried about specific material properties. RAG addresses this limitation by enabling LLMs to dynamically access and incorporate information from external, structured sources [19], such as private EPD datasets. This capability becomes especially crucial when responding to domain-specific queries that require accurate data. For instance, when asked, "*What is the declared reused vs recycled content of steel used in the Astron PRS Product?*", a standalone LLM produces a vague response (Fig. 1), suggesting the use of EPDs without offering a definitive answer. In contrast, the same question posed to an LLM enhanced with RAG yields an accurate response. This is possible because RAG retrieves the relevant data directly from the product's EPD and provides it as context to the LLM. It highlights how combining semantic search with authoritative EPD data enhances the accuracy, specificity, and trustworthiness of AI-generated insights, transforming how technical information is accessed and applied in sustainable construction practices.

Based on a RAG approach with LLMs, this study introduces EcoQuery, a system designed to streamline user interaction with EPDs for structural steel. By integrating intelligent semantic search with NL reasoning (i.e., the ability of AI to understand, interpret, and generate responses based on complex human language queries [17]), EcoQuery empowers users to pose high-level sustainability questions and receive accurate, context-aware answers directly derived from steel EPD data. This capability is crucial for translating complex environmental data into actionable insights without requiring specialised expertise. This innovative approach offers an alternative to traditional EPD analysis workflows by:

(1) Reducing reliance on expert tools and manual parsing, while improving access to comprehensive steel LCA data to enable faster, data-driven decisions during early-stage material selection.
(2) Supporting climate-conscious design practices by providing a user-friendly NL QA interface that promotes the adoption of low-carbon steel solutions in construction.
(3) Ensuring support scalability and sustainability, allowing new EPDs to be easily added or updated based on user requirements with minimal technical knowledge and cost.

The remainder of the paper is organised as follows: Section 1 provides the introduction; Section 2 reviews the background and related work; Section 3 details the EcoQuery system, including the dataset, implementation, and evaluation; Section 4 presents the findings, along with their benefits and limitations; and Section 5 concludes the paper.

2. Background and related work

This section reviews the existing landscape in environmental assessment of structural materials, with a particular focus on structural steel, EPD interpretation challenges, and the recent use of AI techniques, especially RAG-LLM frameworks for diverse applications.

2.1. Environmental impact of structural steels

Structural steel is among the most carbon-intensive materials used in the built environment [20]. Numerous studies have quantified its embodied carbon and life cycle impacts, particularly focusing on metrics such as GWP, ADPF, and recycling potential. These analyses are often supported by LCA methodologies [21] and typically rely on data sourced from EPDs [22].

EPDs are standardised documents developed under ISO 14025 and EN 15804 [9]. They provide comprehensive data on a product's environmental performance throughout its life cycle and are increasingly recognised as essential tools for enabling transparent, data-driven decisions in sustainable construction. With increasing regulatory pressure on the construction sector to reduce its environmental impact, EPDs are becoming essential tools for ensuring compliance and demonstrating accountability [8]. ECO-Platform, a European non-profit association providing standardised access to EPDs, is one of the leading platforms

Table 1
Count of EPDs by country on the ECO-Platform.

Country [Code]	Count of EPDs	Country [Code]	Count of EPDs	Country [Code]	Count of EPDs
United Arab Emirates (AE)	145	Global (GLO)	4306	Poland (PL)	169
Austria (AT)	302	Ireland (IE)	190	Portugal (PT)	120
Australia (AU)	1224	Israel (IL)	122	Rest of Asia (RAS)	18
Belgium (BE)	32	India (IN)	113	Rest of Europe (RER)	4501
Bahrain (BH)	21	Iceland (IS)	47	Rest of Latin America (RLA)	16
Brazil (BR)	42	Italy (IT)	2494	Rest of Middle East (RME)	21
Switzerland (CH)	52	Lithuania (LT)	89	Rest of North America (RNA)	76
Chile (CL)	10	Luxembourg (LU)	13	Romania (RO)	73
China (CN)	377	Latvia (LV)	34	Saudi Arabia (SA)	118
Germany (DE)	1163	Mexico (MX)	25	Scandinavia (SCAND)	13
Denmark (DK)	800	Malaysia (MY)	46	Sweden (SE)	1460
Estonia (EE)	188	Netherlands (NL)	96	Southeast Asia (SEA)	69
Egypt (EG)	40	Norway (NO)	1720	Slovenia (SI)	22
Spain (ES)	270	Nordic Region (NORD)	85	Slovakia (SK)	37
European Union (general) (EU)	173	New Zealand (NZ)	110	Thailand (TH)	25
European Union (EU-27)	93	Oceania (OCE)	25	Turkey (TR)	286
Finland (FI)	380	Oman (OM)	41	United Kingdom (UK)	697
France (FR)	61	Philippines (PH)	32	United States (US)	26

for accessing life cycle data [12]. It hosts a comprehensive collection of approximately 23,828 third-party verified EPDs, all published in accordance with the EN 15804 and ISO 14025 standards. These EPDs are available in either PDF or ILCD+EPD formats and contain detailed LCA data across modules A1–C4 and D, covering stages from raw material extraction to end-of-life and potential benefits. The EPDs span a wide range of countries, including France, the United Kingdom, and many others. Table 1 illustrates the distribution of EPDs across selected countries, highlighting regions with significant activity. Table 2 breaks down the top 24 product domains by the number of EPDs published, drawn from over 330 distinct classifications on the platform. This concentration in construction products, infrastructure, and building materials reflects the growing global adoption of EPDs and sector priorities. The extensive volume and geographic diversity of ECO-Platform's EPD data offer valuable opportunities for regional sustainability analyses, while also revealing potential gaps for future research and development.

Despite their technical robustness, several studies have identified persistent barriers that limit the effective use of EPDs in practice. These challenges include the complexity of interpreting tabular and text-heavy content, insufficient integration into early-stage design workflows, and inconsistent formats that hinder meaningful cross-product comparisons [23,24]. Although EPDs are instrumental in comparative LCA studies, such as those evaluating structural systems made of concrete, steel, or wood, variation in impact reporting remains a significant challenge [23,25]. Importantly, Anderson and Jones [24] argue that much of this variation reflects legitimate differences in production technology, geographic factors, and product-specific characteristics rather than methodological flaws. These findings counter doubts about EPD reliability and emphasise the urgency of using EPD-based assessments now, rather than waiting for perfect data harmonisation. Otero et al. [26] support this view by highlighting significant discrepancies, up to 90% in later life cycle stages, often due to poor traceability and uncertainty quantification. Case studies such as Morris et al. [27], which compare the embodied carbon of glue-laminated timber and steel, and Saufnay and Demonceau [28], demonstrated carbon and cost savings from high-strength steels, further illustrate the strategic value of EPDs in guiding low-carbon material choices. These examples underscore the evolving nature of LCA methodologies.

While challenges in standardisation, accessibility, and interpretation remain, EPDs have already proven valuable for quantifying environmental impacts and supporting carbon-conscious decision-making. As the above-mentioned environmental assessment approaches become more widely adopted and regulations increasingly demand transparency, the integration of EPDs into construction practice will be both necessary and transformative for the sector. Building on this foundation of data-driven sustainability, recent advances in AI, particularly LLMs, offer innovative opportunities to further enhance decision-making processes.

2.2. LLMs with RAG for sustainability

LLMs are advanced AI systems can perform a range of natural language processing (NLP) tasks, including summarisation, translation, text generation, and QA [17]. Their ability to process and synthesise large volumes of information makes them valuable for addressing complex, interdisciplinary challenges such as those found in sustainability domains. LLMs are typically accessed via prompts (i.e., textual instructions that guide the model's output). Depending on the model type, users can adapt prompts for QA tasks such as carbon footprint comparison, EPD interpretation, or lifecycle data extraction and analysis. LLMs are categorised into three broad types [29]:

- *Proprietary LLMs*: Offered via paid APIs or web interfaces, these models generally provide high performance and enterprise support but limit transparency regarding training data.
- *Open LLMs*: Share model architectures and weights publicly, enabling partial reproducibility.
- *Open-source LLMs*: Provide full transparency by releasing training data, code, and weights, promoting reproducibility and community-driven research.

Open-source models provide researchers with accessible, cost-free platforms to customise LLMs, promoting transparency and fostering innovation, both crucial for sustainable AI development. In contrast, proprietary models, while often more powerful and user-friendly, tend to limit openness and come with higher financial costs [29]. Despite their promise, LLMs face significant sustainability challenges: training large models demands substantial energy and computational resources, contributing to carbon emissions. Additionally, LLMs may produce factually incorrect or biased outputs [19], which can mislead sustainability assessments if relied upon uncritically. Their dependence on static training data also limits awareness of recent developments, such as updated material standards. RAG addresses these limitations by combining LLMs with external, dynamic knowledge sources, allowing models to access and incorporate up-to-date, domain-specific information during response generation [19]. This integration enhances factual accuracy, and contextual relevance, qualities essential for informed decision-making across applications [30,31]. By contextualising LLMs with relevant domain knowledge and enabling their reuse in diverse settings, RAG contributes to more sustainable AI solutions.

Prior work demonstrates that RAG enhances LLM accuracy by grounding responses in external domain-specific sources, particularly in healthcare applications [32]. Building on this, BioRAG leverages a

Table 2
Distribution of EPDs across construction-related categories on the ECO Platform.

	Classification of EPDs – Count by Category	
1.	Construction products, Infrastructure & buildings	7708
2.	Construction products	2848
3.	Construction/ Electronic and electrical components and products	1049
4.	Construction products/ Normal- light and aerated concrete/ Concrete components made of in situ or delivered concrete	436
5.	Construction/ Paints	425
6.	Construction/ Technical and chemical construction products	404
7.	Construction/ Insulation	368
8.	Construction/ Ventilation components	363
9.	Furniture/ Seating	344
10.	Construction/ Concrete products	314
11.	Construction/ Steel, aluminium structures	290
12.	Construction/ Pipe systems	230
13.	Construction/ Asphalt and crushed stone	207
14.	Basic materials and preliminary products/ Cement, building lime and other binders/ Cement	204
15.	Furniture & other goods, Construction products	203
16.	Building services/ Electrical/ Lighting	199
17.	Building/ Prefabricated concrete	168
18.	Furniture/ Board	127
19.	Building/ Electrical cables	126
20.	Basic materials and preliminary products/ Cement, building lime and other binders/ Cement	125
21.	Mineral building materials/ Stones and elements/ Bricks	118
22.	Building/ Doors and windows	110
23.	Mineral building materials/ Stones and elements/ Prefabricated concrete elements and concrete products	104
24.	Building products/ Floor coverings/ Floor coverings	95

domain-specific embedding model and hierarchical retrieval of scientific papers to outperform previous models through structured retrieval and stepwise reasoning in life science QA [33]. Similarly, XLR-KGDD combines RAG and fine-tuning to map multimodal clinical data into patient knowledge graphs, enabling precise disease diagnosis alongside guideline-based explanations [34]. TC-RAG further advances this by incorporating a Turing Complete System for managing state variables, resulting in over 7% improvement in medical knowledge retrieval accuracy [35]. Beyond healthcare, RAG-powered LLMs have shown promising results in other fields. For instance, an intelligent financial analysis system significantly improves accuracy when analysing NASDAQ data spanning 2010 to 2023 [36]. In engineering education, an AI-driven multiple-choice question generator employs LLMs with adaptive learning and dual-prompt engineering to tackle STEM retention issues [37]. Similarly, Ask-EDA, a conversational agent for electronic design engineers, uses hybrid RAG and abbreviation de-hallucination techniques, achieving over 40% recall improvement on specialised technical queries [38]. Additionally, an automated system for generating future work suggestions from scientific articles, enhanced with RAG and LLM feedback mechanisms, outperforms traditional baselines in quality and relevance [39]. Lastly, BeefBot, a chatbot tailored for beef producers, integrates RAG and fine-tuning to deliver accurate, customised agricultural advice via a continuously updated, user-friendly, and confidential web platform [40].

Recent studies have begun to explore the use of RAG combined with LLMs in construction environmental management, particularly to address data fragmentation, interoperability, and decision-making challenges. Study [41] proposes an intelligent embodied carbon accounting framework for buildings that integrates LLMs with a three-stage closed-loop RAG architecture. By grounding model reasoning in retrieved life-cycle data and standards, the framework improves calculation reliability and efficiency, achieving an approximate 25% increase in embodied carbon assessment accuracy. Study [42] presents an open-access AI-driven tool that semi-automatically matches BIM construction elements with environmental LCA datasets, targeting one of the most labour-intensive steps in sustainability assessment. Using a Python-based workflow combined with a Generative Pre-trained Transformer (GPT)-powered matching mechanism, the approach enhances automation, interoperability, and early-stage environmental evaluation in construction planning. Extending RAG–LLM applications to material engineering, study [43] introduces a RAG-based dam concrete mix proportioning system that integrates knowledge graphs with semantic retrieval to generate mix designs optimised for strength, durability, and crack resistance. Validation results demonstrate that the system outperforms larger standalone LLMs while providing accurate and interpretable recommendations. Finally, study [44] develops an AI-powered Sustainable Building Bot that leverages Generative AI (GenAI) to support regulatory intelligence and sustainability assessment in smart city construction, reporting improvements in efficiency and accuracy compared to manual workflows, while emphasising the continued importance of human oversight for contextual and ethical reliability. These diverse applications illustrate RAG's potential to enrich LLM knowledge and improve domain-specific QA tasks. However, despite this progress, the use of RAG with LLMs for environmental impact analysis remains limited. Given RAG's ability to integrate multiple external datasets, such as EPDs relevant to sustainability assessments, there is significant potential to advance this field. In the next section, EcoQuery is introduced, to enable NL querying of environmental performance data.

3. Proposed system

EcoQuery, the AI assistant developed in this study (Fig. 2), builds upon a RAG framework integrated with a LLM to enable intelligent NL querying of environmental performance data derived from EPDs. While RAG has been widely adopted for domain-specific QA, EcoQuery distinguishes itself from standard RAG pipelines through a domain-aware system design tailored specifically to the structure, semantics, and regulatory context of EPD-based LCA data. In particular, the proposed approach introduces lifecycle-aware data chunking aligned with EN 15804 modules (A1–A3, C1–C4, and D), metadata-constrained semantic retrieval across material type, geography, and publication year, and prompt strategies that enforce quantitative, stage-consistent, and non-hallucinatory responses. These design choices enable structured environmental reasoning tasks, such as cross-stage impact comparison, interpretation of recycling credits, and comparative product ranking, that are not supported by generic RAG-based systems.

For the proof-of-concept demonstration, the Gemini LLM (developed by Google) [45,46] was selected due to its accessibility and adaptability. The system consists of several interconnected modules designed to convert structured and semi-structured technical data into accurate and user-friendly responses. The following sections present the full pipeline, from dataset construction to information retrieval and LLM-based answer generation, explaining how each component contributes to producing domain-specific insights. The final section describes the evaluation of the system's performance.

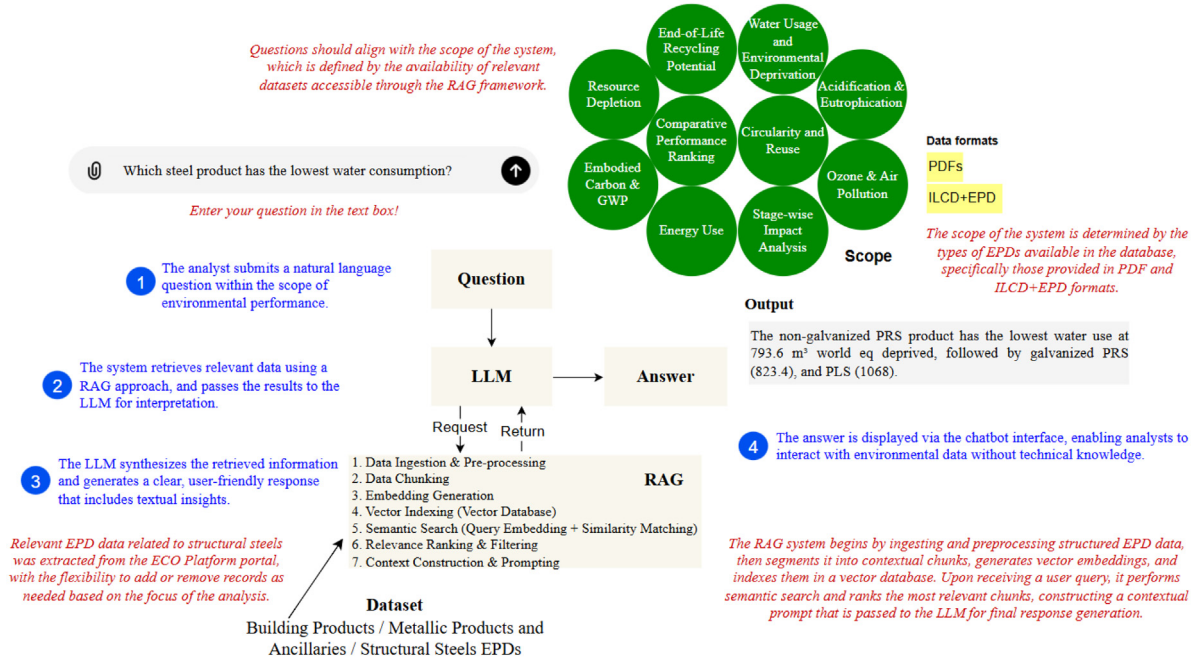


Fig. 2. System workflow for RAG-based querying of EPDs.

Table 3
Key environmental impact topics and their relevance to the ecoquery system for structural steel analysis.

No.	Topic	Relevance
1	Embodied Carbon & GWP	Core metric extracted and compared for low-carbon material selection.
2	Resource Depletion	To assess fossil and non-fossil resource impacts in line with comprehensive LCA analysis.
3	End-of-Life Recycling Potential	Supports circular economy evaluation via recycling and Module D credits.
4	Energy Use	Covers lifecycle energy demand for full environmental profiling.
5	Water Usage and Environmental Deprivation	Included for holistic sustainability assessment.
6	Acidification & Eutrophication	Provides insight into regional pollution and ecosystem impacts.
7	Ozone & Air Pollution (ODP/POCP)	Enables air quality impact assessment within lifecycle analysis.
8	Stage-wise Impact Analysis	Facilitates detailed lifecycle phase attribution (A1–D).
9	Circularity and Reuse	Supports analysis of reuse, recycling, and end-of-life scenarios for circular economy.
10	Comparative Performance Ranking	Enables material ranking and informed decision-making.

3.1. Dataset overview

The first step in implementing EcoQuery involves selecting and preparing an appropriate dataset. For this study, a focused subset of 12 EPDs was curated from the ECO-Platform, given the pivotal role of structural steel in evaluating environmental performance in structural design. This dataset is intended as a proof-of-concept, designed to demonstrate the system's capabilities rather than to exhaustively represent all available steel EPDs. The EPDs were primarily sourced from manufacturers based in Germany (DE), the European Union (EU-27), and Rest-of-Europe (RER) regions to ensure regional relevance while maintaining consistency in environmental indicators and reporting conventions. Each EPD entry includes structured metadata such as classification type, version ID, country of origin, publication year, registration number, Universally Unique Identifier (UUID), and Uniform Resource Identifier (URI) link, as well as declared unit, system boundaries, life-cycle modules (A1–A3, C1–C4, and Module D), and product classification schemes. These metadata fields are retained during ingestion and used to support metadata-constrained retrieval and life-cycle–aware query filtering. These EPDs support the EcoQuery system in addressing a wide range of sustainability topics (see Table 3 for how each topic is mapped within the EcoQuery system to support environmental impact analysis of structural steels). These topics form the basis for domain-specific QA in EcoQuery, offering users interpretable and actionable insights. The selection of steel-related EPDs and their associated topics in this study is intended for demonstration purposes; users can customise their analysis by selecting relevant EPDs to adjust the scope according to their specific needs. The next section outlines the steps involved in implementing the EcoQuery AI assistant using the EPD data.

3.2. System implementation

The implementation process began with downloading a selected set of EPDs related to structural steels from the ECO-Platform platform. As the EPD data was already clean, no additional preprocessing techniques were necessary. To enable RAG functionality, the first step involved chunking the retrieved data (Table 4). Chunking refers to dividing the structured EPD content into smaller, meaningful segments to improve information retrieval. In this study, the data was segmented into chunks based on product types and their associated life cycle stages (e.g., A1–A3, C1–C4, and Module D), with each chunk containing approximately 300–500 tokens. This logical segmentation enhances retrievability by allowing the system to identify and retrieve the most relevant sections of data for answering user queries accurately. For example, one chunk may include the impacts from stages A1 to A3, while another contains information on Module D recycling benefits or end-of-life emissions. Each chunk is enriched with contextual metadata such as product type, country of origin, year of publication, and material classification. Metadata-based filtering is applied prior to similarity search to constrain retrieval to contextually relevant EPD entries. As the data is chunked, each unit is transformed into a dense vector representation using a transformer-based embedding model. The Sentence-BERT embedding model [47] is used to capture the semantic meaning of each

Table 4
RAG implementation parameters for EPD-based QA.

No.	Component	Parameter/ Setting
1	Chunking strategy	Semantic chunking by product type and life-cycle module (A1–A3, C1–C4, D).
2	Chunk size	~300–500 tokens
3	Embedding model	Sentence-BERT
4	Vector database	FAISS
5	Similarity metric	Cosine similarity
6	Retrieval strategy	Top-k nearest neighbour search (k = 5).
7	Metadata filtering	Material category, country, publication year, life-cycle stage.
8	Context assembly	Retrieved chunks concatenated and truncated to fit LLM context window.

Table 5
Prompt design for EPD-based QA with RAG-LLM approach.

1. System Prompt (Instructions to the LLM)

You are a sustainability analyst specialising in environmental impact analysis of structural steel using EPDs. Your task is to answer technical questions using only the provided EPD context. Extract and compare environmental indicators such as:
- Global Warming Potential (GWP)
- Abiotic Depletion Potential of Fossil Fuels (ADPF)
- Recycled content or recovery rates
- Life cycle stage data (A1–A3, A4, C1–C4, D)

Be concise, quantitative, and reference exact values from the context. If multiple products are mentioned, make clear comparisons. If the answer cannot be found in the context, respond with: "*Insufficient data to answer*."

2. User Prompt Template

Question:
{user_question}

Context:
{retrieved_epd_chunks}

3. Expected LLM Behaviour

- Ground the answer strictly in the provided context.
- Reference product names and dataset numbers where possible.
- Return short, informative answers (1–3 sentences max).
- Do not hallucinate or generalise beyond the given data.

4. Example Prompt

System Prompt: (as above)
User Input:
Question: Which structural steel product has the lowest total GWP?
Context:
EPD Dataset 1: PRS (galvanised): GWP-total = 2553 kg CO_2-eq (A1–A3)
EPD Dataset 2: PLS: GWP-total = 2277 kg CO_2-eq (A1–A3)
EPD Dataset 4: Hot rolled coil: GWP-total = 2082 kg CO_2-eq (A1–A3)
Expected Answer:
The hot rolled steel coil (EPD Dataset 4) has the lowest total GWP of 2082 kg CO_2-eq in the A1–A3 stage.

chunk beyond simple keyword matching. The resulting embeddings are stored in a scalable FAISS vector database [48], indexed using cosine similarity to support efficient approximate nearest-neighbour search. During query time, a top-k semantic retrieval strategy (k = 5) is employed to identify the most relevant chunks, which are then passed to the LLM as contextual input for grounded answer generation.

When a user submits a NL query, for example, "*Which steel product has the lowest water consumption?*", the system first embeds the query into the same vector space as the stored chunks. A similarity search is then conducted to retrieve the most semantically relevant EPD chunks. A ranking algorithm evaluates the retrieved results based on their semantic similarity to the query and their metadata alignment. The top-ranked chunks are compiled into a contextual prompt that falls within the token limits of the LLM. To ensure accurate responses from the LLM, a four-part prompt design strategy is used (Table 5). First, the system prompt defines the LLM's role as a *sustainability analyst* tasked with answering technical questions strictly based on retrieved EPD data. Second, the user prompt template combines a NL query with semantically relevant EPD chunks. Third, explicit behavioural expectations guide the model to provide concise, quantitative responses, referencing specific product names and dataset identifiers while avoiding hallucination. Finally, an example prompt and expected answer is included to demonstrate the desired output format and reasoning process, helping the model understand how to extract and compare values such as GWP and ADPF. This comprehensive prompt structure supports consistent, interpretable results across diverse EPD-based queries.

The constructed prompt is passed to the Gemini LLM, which synthesises the retrieved content and generates a NL response. Gemini can perform multi-step reasoning [45,46], including comparative analysis across products, numerical evaluation (e.g., *identifying the lowest environmental impact*), and explanatory interpretation (e.g., *why one product is more sustainable than another?*). The output is structured to be clear, often including both textual summaries and numerical highlights. This makes the information accessible to both technical and non-technical users, supporting a broad range of decision-making contexts. Finally, the generated response is delivered through a chatbot-style user interface. This interface allows users to interact with the system intuitively, without requiring any technical expertise in LCA or data formats. Users can submit follow-up questions, refine their queries based on specific countries or materials, and retrieve answers in text formats. This front-end design ensures that sustainability analysts, architects, engineers, and policy makers can easily access and interpret complex EPD data, effectively bridging the gap between environmental datasets and practical, real-world decision-making.

After implementing the system, users can interact with EcoQuery to explore a wide range of sustainability topics (Table 3 and Fig. 2) through NL queries. The AI assistant enables users to access and interpret complex environmental performance data in an intuitive and meaningful way. Table 6 presents a selected set of diverse questions and corresponding answers that demonstrate the practical functionality and domain relevance of EcoQuery.

Table 6
A selected set of diverse questions demonstrating the functionality of EcoQuery.

No.	Topic	Question
1	**Embodied Carbon & Global Warming Potential (GWP)** *Evaluate and compare the total lifecycle carbon emissions of materials.*	**Q1.** Which structural steel product has the lowest total GWP across its life cycle? **Q2.** How much embodied carbon can be saved by switching from PRS (galvanised) to PLS elements?
2	**Resource Depletion** *Impact on fossil and non-fossil resources.*	**Q3.** Which product shows the highest fossil resource depletion (ADPF)? **Q4.** How does non-fossil resource depletion differ between PRS and PLS?
3	**End-of-Life Recycling Potential** *Understand environmental credits and circularity.*	**Q5.** Which product offers the greatest recycling benefit in Module D? **Q6.** How do the recycling benefits compare between hot rolled steel and PRS?
4	**Water Usage and Environmental Deprivation** *Assess water use impacts.*	**Q7.** Which structural steel product has the highest water use? **Q8.** How does water use differ between PLS and PRS?
5	**Energy Use** *Evaluate energy demand across the lifecycle.*	**Q9.** Which product consumes the most energy (ADPF) during production? **Q10.** How much energy is saved from recycling in hot rolled steel?
6	**Acidification & Eutrophication** *Analyse regional pollution and ecosystem impact.*	**Q11.** Which product contributes most to acidification? **Q12.** What's the difference in freshwater eutrophication potential between PRS and PLS?
7	**Ozone & Air Pollution** *Evaluate air quality-related impacts.*	**Q13.** Which material has the lowest POCP? **Q14.** What is the total ODP for hot rolled steel?
8	**Stage-wise Impact Analysis** *Life cycle phase contribution.*	**Q15.** Which stage contributes the most to GWP in PRS? **Q16.** What is the role of installation (A5) in total emissions?
9	**Circularity and Reuse** *Evaluate reuse and landfill.*	**Q17.** How much steel is reused vs recycled in PRS? **Q18.** Which scenario shows 100% reuse in hot rolled steel?
10	**Comparative Performance Ranking** *Best performer identification.*	**Q19.** Which product has the best overall environmental profile? **Q20.** Which product shows the greatest benefit from Module D credits?

3.3. System evaluation

To evaluate the performance of EcoQuery, a structured assessment was conducted using 60 domain-specific queries manually constructed from the content of the selected structural steel EPDs. The questions were designed to reflect typical practitioner-oriented information needs and to ensure coverage across multiple EPD types, environmental indicators, and life-cycle stages. Specifically, the query set spans key sustainability topics such as global warming potential, resource and energy use, pollution, water consumption, and material circularity, and includes questions targeting production (A1–A3), transport and installation (A4–A5), use and end-of-life stages (C1–C4), as well as Module D benefits. Reference answers were manually derived from the original EPD documents by extracting the corresponding numerical values, units, and life-cycle stage information directly from the reported tables and textual descriptions. These reference answers were cross-checked against the source EPDs to ensure consistency and correctness prior to evaluation. Each response generated by EcoQuery was analysed based on critical information components, including numerical values, units, referenced life-cycle stages (e.g., A1–D), and environmental indicators (e.g., ADPF, POCP). While various evaluation metrics exist for QA systems [49], a focused set was selected for this study to align with the structured and technical nature of EPD data. These include Accuracy, Precision, Recall, Numeric Exact Match (EM), and BERTScore [50]. Together, these metrics capture complementary aspects of system performance, ranging from strict numerical correctness to semantic fidelity, which are essential for reliable EPD-based decision support. Table 7 presents the definitions, benefits, and limitations of each metric.

EcoQuery demonstrated strong performance across key evaluation metrics (Table 8). It achieved an accuracy of 0.83, meaning 83% of its responses included all critical components such as values, stages, and units. With a precision of 0.87 and recall of 0.85, the system consistently delivered factually correct and comprehensive answers, even in multi-part queries. The numeric exact match (0.81) confirmed reliable numerical accuracy, with only minor rounding or unit variations. Lastly, a BERTScore of 0.89 indicated high semantic similarity between generated and reference answers, reflecting EcoQuery's strong understanding of environmental data language. In practical terms, these results indicate that EcoQuery can be reliably used to extract and compare quantitative EPD indicators for early-stage sustainability decision-making, with most observed discrepancies being minor and unlikely to affect high-level material selection or hotspot identification. To further contextualise these results, a qualitative error analysis was conducted. A review of incorrect or partially correct responses indicates that most errors arise in queries requiring simultaneous reasoning across multiple environmental indicators or life-cycle stages, particularly when information is sparsely reported or inconsistently structured within the source EPDs. Minor discrepancies were also observed in numerical exact match, primarily due to rounding differences or unit normalisation rather than incorrect value retrieval. EcoQuery performs particularly well on queries involving single-indicator extraction (e.g., GWP, ADPF), stage-specific impact analysis, and comparative ranking of products within the same life-cycle scope. In contrast, queries that combine multiple indicators, require implicit assumptions, or depend on non-declared modules were more challenging and often resulted in incomplete responses. These findings highlight EcoQuery's strength in structured, data-grounded EPD interpretation, while indicating areas where enhanced reasoning control and uncertainty handling could further improve performance.

4. Discussion

This study introduced EcoQuery, a RAG–LLM-based system designed to transform how EPDs are accessed and interpreted for structural steels. Structural steel is highlighted in this study not because it is unique in having EPDs, but because it represents one of the most carbon-intensive and decision-critical materials in building structures. While all materials listed in Table 2 follow the same standardised EPD framework (ISO 14025 and EN 15804), steel differs in practice due to its high embodied carbon in early life-cycle stages (A1–A3), substantial mass contribution at the building scale, and complex end-of-life modelling involving reuse and recycling credits in Module D. These characteristics introduce greater variability and interpretive complexity compared to many other construction materials, even under an identical EPD format. In addition, structural steel EPDs are selected for the

Table 7
Evaluation metrics for different types of EPD-based QA questions.

No.	Metric	Definition	Benefits	Limitations
1	Accuracy	Fraction of queries with all correct components (e.g., correct product, numerical value, life cycle stage, interpretation). Binary scoring: 1 if fully correct, 0 otherwise.	Captures complete correctness rather than partial answers.	Requires clear definition of "correct components". No credit for partial correctness.
2	Precision	Measures how exact the extracted facts are.	Useful for multi-fact answers, gives partial credit for correct components.	Less effective for qualitative facts.
3	Recall	Measures completeness of fact extraction.	Captures whether all necessary facts are found. Valuable when completeness is critical (e.g., multiple facts expected).	Requires manual labelling.
4	Numeric EM	Strict correctness checks on numbers only.	Highly precise for technical tasks where numeric correctness matters. Penalises numeric errors that semantic metrics miss.	No partial credit for close or approximate numbers. Not useful for non-numeric content.
5	BERTScore	Semantic similarity score between predicted and reference texts using contextual embeddings.	Captures meaning similarity beyond exact word matches. Handles paraphrasing and minor variations.	Less sensitive to numeric errors. Can be overly generous with semantically similar but factually incorrect answers.

Table 8
Performance evaluation of ecoquery across selected metrics.

No.	Metric	Score	Interpretation
1	Accuracy	0.83	Majority of the answers included all key components correctly (e.g., values, stages, units, etc.).
2	Precision	0.87	High correctness in factual responses.
3	Recall	0.85	Most relevant facts were retrieved, with few omissions in multi-part answers.
4	Numeric EM	0.81	Good numerical accuracy, though some responses had minor rounding or unit issues.
5	BERTScore	0.89	Strong semantic similarity, with answers generally close in meaning to references.

sake of demonstration, as they provide a data-rich and high-impact use case to illustrate the capabilities of the proposed RAG–LLM framework, which remains transferable to other material categories.

The proposed system directly addresses critical limitations in traditional EPD workflows, namely reduced accessibility, reliance on expert LCA knowledge, and the complexity of semi-structured data formats, by enabling NL QA over environmental datasets. EcoQuery delivers on the three core objectives outlined in this study (see Table 9). First, it replaces manual parsing of PDF-based EPDs with automated information extraction [51] using RAG, significantly reducing the need for expert interpretation. Traditional workflows often involve manually reviewing dense technical documents and modelling environmental impacts using custom scripts or spreadsheets. In contrast, EcoQuery allows users to query key sustainability metrics such as GWP, ADPF, and recycling potential using plain English. This automated extraction is coupled with plain language translation of technical indicators, enabling non-expert stakeholders to engage meaningfully with environmental data during early-stage material selection, an essential phase for reducing embodied carbon in buildings. Second, EcoQuery supports carbon-conscious design by offering features that go well beyond static data interpretation. For instance, users can ask open-ended questions like, "*Which life cycle stage contributes the most to CO_2 emissions?*" or "*Which steel product has a lower ADPF than Astron PLS?*", queries that would otherwise require custom LCA tools or domain expertise. EcoQuery also supports scenario simulation, such as evaluating different end-of-life processing routes or exploring reuse vs. recycling strategies, thereby enhancing decision-making flexibility. These features enable hotspot analysis, scenario exploration, and comparative material assessment without needing extensive manual modelling. By surfacing actionable insights from EPDs, the system embeds sustainability directly into architectural, engineering, and procurement workflows.

Third, scalability and sustainability are fundamental design principles embedded within the system's architecture. The use of effective chunking and metadata tagging methods during data ingestion allows new EPDs to be incorporated seamlessly with minimal manual effort. This modular approach ensures that EcoQuery remains highly adaptable to evolving datasets, enabling continuous expansion as new EPD sources emerge and existing ones are updated. While the current implementation focuses on structural steel EPDs, the system's flexible framework supports integration of EPDs from other material categories and regions. At the same time, cross-material and cross-regional integration introduces technical challenges, including differences in functional units, production assumptions, background datasets, and end-of-life modelling conventions, which require careful metadata harmonisation and consistent life-cycle boundary interpretation. Importantly, the value of the proposed approach is not dependent on the availability of large numbers of EPDs for a single project. In practice, construction projects often face constrained material choices due to structural, regulatory, or supply-chain limitations. Even in such cases, EcoQuery provides clear advantages by reducing the cognitive and technical burden of interpreting EPD assumptions, enabling structured reasoning across life-cycle stages, and supporting scenario-based queries (e.g., reuse versus recycling) that are difficult to perform reliably through manual review. Additionally, EcoQuery's architecture is not limited to EPDs alone; it can incorporate other relevant textual datasets, such as sustainability reports, regulatory documents, or industry guidelines, to enhance environmental impact analyses. This extensibility increases the system's value by providing more comprehensive sustainability insights beyond conventional EPD data. However, incorporating larger and more diverse datasets may affect retrieval performance, potentially increase response times, or impact accuracy. Addressing these challenges represents an important direction for future research, including optimising retrieval algorithms, improving indexing strategies, and balancing dataset breadth with system responsiveness to ensure a high-quality user experience.

While EcoQuery shows potential as a domain-aware RAG–LLM system for EPD-based environmental analysis, several limitations remain and point to important directions for future research:

1. *Single-LLM dependency*: The current implementation relies on a single LLM from the Gemini family. Comparative evaluation across multiple LLM architectures (e.g., open-source and proprietary models) was not conducted and could reveal performance differences in handling complex, multi-dimensional sustainability queries.

Table 9
EcoQuery AI assistant benefits over traditional workflow for EPD-based sustainability analysis.

No.	Feature	Traditional workflow	EcoQuery system
1	Data Extraction	Manual reading/parsing of EPD PDFs.	Automated retrieval using RAG from structured/unstructured EPDs.
2	Impact Interpretation	Requires LCA expertise to interpret environmental metrics.	Converts technical indicators into plain language for non-experts.
3	Carbon Hotspot Analysis	Spreadsheet modelling and manual lookup.	NL query (e.g., "Which stage emits the most CO_2?").
4	Scenario Exploration	Requires custom LCA tools or scripts.	LLM supports reasoning over alternative scenarios based on declared EPD data.
5	Material Comparison	Manual cross-check across databases like EcoPlatform or manufacturer EPDs.	Instant comparative analysis (e.g., "Compare GWP of Astron PLS and PregyMetal").
6	Communication	Dense technical tables not suitable for all audiences.	Auto-generated summaries, tables, visuals, and executive reports.
7	Data Integration	Requires harmonising formats (PDFs, CSVs, etc.) manually.	Multimodal data ingestion across text, tables, and metadata.
8	Cross-stage Reasoning	Difficult to track inter-stage relationships (e.g., reuse affects A1–D).	LLM enables inter-stage reasoning and explains trade-offs (e.g., "What if no reuse?").
9	Standard Compliance Checking	Needs human review of EPD conformance to ISO 14025/EN 15804.	Can check EPD structure against standards and flag inconsistencies.
10	Personalised Queries	Analysts must filter through all data to answer bespoke queries.	Tailored Q&A: E.g., "How does Astron PLS perform under EU Taxonomy criteria?".

2. *Absence of baseline comparison*: No direct baseline comparison with existing EPD query tools or benchmark systems was performed. This is primarily due to the fact that most existing EPD tools rely on rule-based systems, databases, or conventional LCA software rather than LLM- or RAG-based approaches, limiting the feasibility of meaningful like-for-like comparison.
3. *Limited dataset scale*: The evaluation was conducted on a curated subset of structural steel EPDs and a limited number of manually constructed queries. Although these queries were designed to cover key environmental indicators and life-cycle stages, broader datasets and larger query sets would be required to assess scalability and generalisation.
4. *Lack of expert-in-the-loop validation*: The study did not incorporate formal validation by LCA experts or environmental practitioners. Expert review could improve the assessment of answer correctness, interpretability, and decision relevance, particularly for complex trade-off and scenario-based questions.
5. *User experience evaluation not included*: EcoQuery is currently implemented as a prototype system with a command-line interface. As a result, extended evaluation of user experience, usability, and workflow integration was not conducted. Future versions should incorporate user-facing interfaces and structured usability studies.
6. *Potential bias and uncertainty in retrieval and generation*: Mechanisms for detecting retrieval bias, uncertainty, or incomplete context coverage were not explicitly implemented. For multi-stage or multi-indicator queries, this may result in partial or misleading responses. Future work should focus on bias detection, uncertainty estimation, and improved transparency in the QA pipeline.

Addressing these limitations will strengthen the robustness, scalability, and usability of EcoQuery and support its evolution into a more comprehensive decision-support tool for EPD-based sustainability analysis.

5. Conclusions

This study presented EcoQuery, an AI assistant that integrates RAG with LLMs to enable NL interaction with EPDs for structural steels. By combining semantic retrieval with contextual reasoning, EcoQuery bridges the gap between structured environmental datasets and accessible, data-driven sustainability insights. The system illustrates how verified life-cycle information can be transformed into intuitive, QA formats, empowering designers and engineers to make carbon-conscious material choices grounded in reliable evidence. Methodologically, EcoQuery establishes a transferable framework for linking domain-specific databases with advanced language reasoning, setting a precedent for the responsible application of AI in sustainability research and practice. Its implementation using structural steel EPDs from the ECO-Platform demonstrates the potential of standardised environmental data to support low-carbon design decisions. Moreover, the approach is inherently scalable, other repositories, such as OpenLCA Nexus, can be integrated to enhance data diversity and regional coverage. Achieving interoperability among these databases will be critical for building a globally consistent infrastructure for AI-enabled environmental impact assessment. Beyond the construction sector, EcoQuery exemplifies a broader shift towards green AI and sustainable digital technology. The RAG framework leverages existing LLMs rather than retraining new models, significantly reducing computational and energy demands. This resource-efficient design shows that environmental responsibility can extend to the AI systems themselves. EcoQuery therefore advances sustainability on two interconnected fronts: enabling low-carbon material decision-making and demonstrating energy-efficient, low-impact AI deployment.

The implications of this work are wide-ranging. For researchers, EcoQuery provides a reproducible framework for integrating structured domain data with AI-driven reasoning. For industry practitioners, it simplifies the use of EPDs in procurement, design, and compliance workflows, aligning with ISO 14025 and EN 15804 standards. For policymakers, it offers a digital tool for material benchmarking and embodied carbon reporting. Collectively, these applications highlight the potential of RAG–LLM systems to accelerate decarbonisation across the built environment. Future research should focus on expanding EcoQuery to include additional materials, datasets, and regions, as well as evaluating alternative LLM architectures for enhanced robustness and domain accuracy. Developing mechanisms for uncertainty estimation, bias detection, and explainability will further strengthen system reliability and user trust. In summary, EcoQuery embodies the convergence of sustainable technology and environmental intelligence, offering a pathway towards transparent, AI-assisted decision-making in construction and beyond. By uniting data accessibility, green computation, and low-carbon design thinking, it represents a meaningful step towards a digitally enabled and environmentally responsible built environment.

CRediT authorship contribution statement

Muhammad Arslan: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Saba Munawar:** Writing – review & editing, Methodology, Data curation.

Declaration of Generative AI and AI-Assisted Technologies

Gemini was used in the EcoQuery system, while GPT-4 was used exclusively for language editing purposes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the University of the West of England, Bristol, which made this research possible.

Data availability

The data supporting this study is available for download from the ECO-Platform (https://www.eco-platform.org/home.html; https://eco-portal.eco-platform.org/).

References

[1] World Green Building Council, Embodied carbon, 2025, https://worldgbc.org/climate-action/embodied-carbon/.
[2] J. Min, G. Yan, A.M. Abed, S. Elattar, M.A. Khadimallah, A. Jan, H.E. Ali, The effect of carbon dioxide emissions on the building energy efficiency, Fuel 326 (2022) 124842.
[3] S. Kumar, A. Gangotra, M. Barnard, Towards a net zero cement: Strategic policies and systems thinking for a low-carbon future, Curr. Sustain./Renew. Energy Rep. 12 (1) (2025) 5.
[4] A.N. Conejo, J.P. Birat, A. Dutta, A review of the current environmental challenges of the steel industry and its value chain, J. Environ. Manag. 259 (2020) 109782.
[5] Y. Khalid, M. Wu, A. Silaen, F. Martinez, T. Okosun, B. Worl, et al., Oxygen enrichment combustion to reduce fossil energy consumption and emissions in hot rolling steel production, J. Clean. Prod. 320 (2021) 128714.
[6] F. Maghool, A. Arulrajah, Y.J. Du, S. Horpibulsuk, A. Chinkulkijniwat, Environmental impacts of utilizing waste steel slag aggregates as recycled road construction materials, Clean Technol. Environ. Policy 19 (4) (2017) 949–958.
[7] G. Wang, S. Zeng, B. Xia, G. Wu, D. Xia, Influence of financial conditions on the environmental information disclosure of construction firms, J. Manag. Eng. 38 (1) (2022) 04021078.
[8] Österblad M., Implementation of environmental product declarations into the designprocess: a case study of steel and concrete structures using tekla structures, 2024.
[9] ISO, ISO 14025:2006, Environmental labels and declarations — Type III environmental declarations — Principles and procedures, 2006, https://www.iso.org/standard/38131.html.
[10] Juarez Roberto Ivan Cruz, Stephen Finnegan, The environmental impact of cement production in Europe: A holistic review of existing EPDs, Clean. Environ. Syst. 3 (2021) 100053.
[11] K. Safari, S. Ghandehariun, An integrative BIM-based life cycle assessment framework to address uncertainties: a mixed-methods study under real construction project context, Int. J. Constr. Manag. (2025) 1–22.
[12] ECO-Platform, Welcome to eco platform, 2025, https://www.eco-platform.org/home.html.
[13] O. Olanrewaju, Data Quality Assurance in Construction Environmental Product Declarations for Whole Building Life Cycle Assessment (Doctoral dissertation), Open Access Te Herenga Waka-Victoria University of Wellington, 2025.
[14] A. Aragón, M.G. Alberti, Limitations of machine-interpretability of digital EPDs used for a BIM-based sustainability assessment of construction assets, J. Build. Eng. 96 (2024) 110418.
[15] O.I. Olanrewaju, W.I. Enegbuma, M. Donn, O.N. Oyefusi, Assessment of environmental product declaration and databases: Towards ensuring data quality assurance practices, Environ. Impact Assess. Rev. 112 (2025) 107803.
[16] M. Schmitt, Automated machine learning: AI-driven decision making in business analytics, Intell. Syst. Appl. 18 (2023) 200188.
[17] Z. Luo, Q. Xie, S. Ananiadou, Factual consistency evaluation of summarization in the era of large language models, Expert Syst. Appl. 254 (2024) 124456.
[18] T. Zhang, F. Ladhak, E. Durmus, P. Liang, K. McKeown, T.B. Hashimoto, Benchmarking large language models for news summarization, Trans. Assoc. Comput. Linguist. 12 (2024) 39–57.
[19] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, et al., Retrieval-augmented generation for knowledge-intensive nlp tasks, Adv. Neural Inf. Process. Syst. 33 (2020) 9459–9474.
[20] V.J. Gan, C.M. Chan, K.T. Tse, I.M. Lo, J.C. Cheng, A comparative analysis of embodied carbon in high-rise buildings regarding different design parameters, J. Clean. Prod. 161 (2017) 663–675.
[21] F. Rossi, C. De Bernardi, M. Frey, M. Niero, From past critiques to present challenges: A review of LCA approaches and results in the aluminum industry, Waste Manage. 204 (2025) 114900.
[22] M. Salati, A.A. Costa, J.D. Silvestre, A comprehensive review of dynamic life cycle assessment for buildings: Exploring key processes and methodologies, Sustainability 17 (1) (2025) 159.
[23] M.C. Caruso, C. Menna, D. Asprone, A. Prota, LCA-based comparison of the environmental impact of different structural systems, in: IOP Conference Series: Materials Science and Engineering, vol. 442, no. 1, IOP Publishing, 2018, 012010.
[24] J. Anderson, D. Jones, Real and apparent variations in embodied carbon impacts provided in EPD for construction products, in: The Routledge Handbook of Embodied Carbon in the Built Environment, Routledge, 2023, pp. 335–358.
[25] J.H. de Paula Filho, L. Guimarães, J. Guth, L. Marioni, Life-cycle-assessment of purlin systems: the influence of the low carbon high strength steels in CO2eq. emissions, Ce/papers 6 (3–4) (2023) 483–488.
[26] M.S. Otero, T. Garnica, S. Montilla, M. Conde, J.A. Tenorio, EPD as LCA quality input data, 2023, A Critical Analysis.
[27] F. Morris, S. Allen, W. Hawkins, On the embodied carbon of structural timber versus steel, and the influence of LCA methodology, Build. Environ. 206 (2021) 108285.
[28] L. Saufnay, J.F. Demonceau, Economic and environmental assessment of high-strength steel grades, Ce/Papers 6 (3–4) (2023) 527–532.
[29] L.F. Bouchard, L. Peters, Building LLMs for Production: Enhancing LLM Abilities and Reliability with Prompting, Fine-Tuning, and RAG, Towards AI, Inc, 2024.
[30] M. Arslan, L. Mahdjoubi, S. Munawar, Driving sustainable energy transitions with a multi-source RAG-LLM system, Energy Build. 324 (2024) 114827.
[31] M. Arslan, S. Munawar, L. Mahdjoubi, P. Manu, Monitoring indoor environmental conditions in office buildings using a sustainable agentic RAG-LLM system, Energy Build. (2025) 116276.
[32] K.K.Y. Ng, I. Matsuba, P.C. Zhang, RAG in health care: a novel framework for improving communication and decision-making by addressing LLM limitations, NEJM AI 2 (1) (2025) AIra2400380.
[33] C. Wang, Q. Long, M. Xiao, X. Cai, C. Wu, Z. Meng, et al., Biorag: A rag-llm framework for biological question reasoning, 2024, arXiv preprint arXiv:2408.01107.
[34] P. Bedi, A. Thukral, S. Dhiman, XLR-KGDD: leveraging LLM and RAG for knowledge graph-based explainable disease diagnosis using multimodal clinical information, Knowl. Inf. Syst. (2025) 1–21.
[35] X. Jiang, Y. Fang, R. Qiu, H. Zhang, Y. Xu, H. Chen, et al., TC-RAG: Turing-complete rag's case study on medical LLM systems, 2024, arXiv preprint arXiv:2408.09199.
[36] J. Wang, W. Ding, X. Zhu, Financial analysis: Intelligent financial data analysis system based on llm-rag, 2025, arXiv preprint arXiv:2504.06279.
[37] S. Gopi, D. Sreekanth, N. Dehbozorgi, Enhancing engineering education through LLM-driven adaptive quiz generation: A RAG-based approach, in: 2024 IEEE Frontiers in Education Conference, FIE, IEEE, 2024, pp. 1–8.
[38] L. Shi, M. Kazda, B. Sears, N. Shropshire, R. Puri, Ask-EDA: A design assistant empowered by LLM, hybrid RAG and abbreviation de-hallucination, in: 2024 IEEE LLM Aided Design Workshop, LAD, IEEE, 2024, pp. 1–5.
[39] I.A. Azher, M.J. Mokarrama, Z. Guo, S.R. Choudhury, H. Alhoori, Futuregen: Llm-rag approach to generate the future work of scientific article, 2025, arXiv preprint arXiv:2503.16561.
[40] Z. Zhang, C.A. Wilson, R. Hay, Y. Everingham, U. Naseem, BeefBot: Harnessing advanced LLM and RAG techniques for providing scientific and technology solutions to beef producers, in: Proceedings of the 31st International Conference on Computational Linguistics: System Demonstrations, 2025, pp. 54–62.
[41] Y. Zou, R. Zheng, J. Xia, Enhancing embodied carbon calculation in buildings: A retrieval-augmented generation approach with large language models, Buildings 15 (19) (2025) 3449.
[42] D. Petrosa, P. Haverkamp, J.G. Backes, D. Crampen, J. Blankenbach, M. Traverso, Development of a BIM-based AI-driven matching tool for LCA datasets, Discov. Sustain. 6 (1) (2025) 1237.

[43] Y. Shang, Z. Ke, P. Lin, Q. Ren, W. Zhang, X. Wang, et al., Empowering knowledge graphs with hybrid retrieval-augmented generation for the intelligent mix scheme of mass concrete, Case Stud. Constr. Mater. (2025) e04979.

[44] R.W. Wong, B.P. Loo, Using generative AI-powered sustainable building bots for regulatory intelligence and sustainability assessment of construction: Lessons from Hong Kong, J. Urban Technol. (2025) 1–24.

[45] G. Comanici, E. Bieber, M. Schaekermann, I. Pasupat, N. Sachdeva, I. Dhillon, et al., Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025, arXiv preprint arXiv:2507.06261.

[46] G. Team, R. Anil, S. Borgeaud, J.B. Alayrac, J. Yu, R. Soricut, et al., Gemini: a family of highly capable multimodal models, 2023, arXiv preprint arXiv:2312.11805.

[47] N. Reimers, I. Gurevych, Sentence-bert: Sentence embeddings using siamese bert-networks, 2019, arXiv preprint arXiv:1908.10084.

[48] M. Douze, A. Guzhva, C. Deng, J. Johnson, G. Szilvasy, P.E. Mazaré, et al., The faiss library, 2024, arXiv preprint arXiv:2401.08281.

[49] Y.J. Chang, T.Y. Yu, C.H. Chang, Evaluating the performance of open-source LLMs in local RAG systems: A practical study on low-carbon data applications, in: International Conference on Machine Learning and Soft Computing, Springer, Singapore, 2025, pp. 55–66.

[50] J. Devlin, M.W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 2019, pp. 4171–4186.

[51] H. Liu, Z. Fu, M. Ding, R. Ning, C. Zhang, X. Liu, Y. Zhang, Logical reasoning in large language models: A survey, 2025, arXiv preprint arXiv:2502.09100.